An Exact Band–Gap Atlas for the Delta-Comb Kronig–Penney Hamiltonian

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Abstract

We realize the one-dimensional periodic delta comb by a closed quadratic form for every real coupling and compute its monodromy discriminant without a formal-potential shortcut. Floquet reduction gives an exact spectral criterion, pure absolute continuity, and a complete negative-energy atlas; the coupling $ga = -4$ is the precise zero-energy threshold. This is the operator and discriminant owner. All positive bands and gaps are then indexed on both sides of every Bragg point. Every nonzero coupling opens every Bragg gap, with an explicit high-energy width expansion, and the integrated and ordinary densities of states are obtained with their edge conventions.

1 Closed form, matching owner, and Floquet discriminant

Fix a period $a > 0$ and a coupling $g \in \mathbb{R}$. On $L^2(\mathbb{R})$ define

$$\mathfrak{h}_{a,g}[u] = \int_{\mathbb{R}} |u'(x)|^2\,dx + g\sum_{n\in\mathbb{Z}} |u(na)|^2, \qquad \operatorname{dom}\mathfrak{h}_{a,g} = H^1(\mathbb{R}). \tag{1}$$

The periodized trace inequality makes the sum finite and infinitesimally form bounded relative to the kinetic term. Thus (1) is closed and lower semibounded for either sign of g and defines a unique self-adjoint operator $H_{a,g}$.

Proposition 1 (Operator and monodromy)**.** *The operator acts as $-u''$ off $a\mathbb{Z}$ and has domain characterized by $u \in H^1(\mathbb{R})$, cellwise H^2 regularity with $u'' \in L^2(\mathbb{R}\setminus a\mathbb{Z})$, and*

$$u'(na+) - u'(na-) = gu(na). \tag{2}$$

For $E = k^2 > 0$, propagation from one right limit to the next is conjugate to the determinant-one matrix

$$M(E) = \begin{pmatrix}1 & 0\\ g & 1\end{pmatrix}\begin{pmatrix}\cos(ka) & \sin(ka)/k\\ -k\sin(ka) & \cos(ka)\end{pmatrix}. \tag{3}$$

Its half trace is

$$\Delta(E) = \cos(ka) + \frac{g}{2k}\sin(ka). \tag{4}$$

At $E = 0$ this means $\Delta(0) = 1 + ga/2$ by continuity. At $E = -\kappa^2 < 0$ it means

$$\Delta(-\kappa^2) = \cosh(\kappa a) + \frac{g}{2\kappa}\sinh(\kappa a). \tag{5}$$

Proof. On every centered cell I_n the one-dimensional trace estimate gives, for each $\varepsilon > 0$,

$$|u(na)|^2 \le \varepsilon\|u'\|^2_{L^2(I_n)} + C_{a,\varepsilon}\|u\|^2_{L^2(I_n)}.$$

Summing the disjoint-cell estimates proves the asserted form bound; choosing ε after g proves lower semiboundedness and closedness by the form perturbation theorem. Integration by parts identifies the associated operator and (2). Free propagation followed by the jump gives (3); direct multiplication gives determinant one and (4). Replacing k by $i\kappa$ gives (5), including the displayed limit at zero. □

The Floquet transform decomposes $H_{a,g}$ over $\theta \in [-\pi, \pi)$ into compact-resolvent fibres on $[-a/2, a/2]$. The fibre functions are quasi-periodic at the endpoints and obey (2) at the origin. A fibre eigenvalue satisfies

$$\Delta(E) = \cos\theta. \tag{6}$$

Theorem 2 (Spectral type and multiplicity)**.** *For every $a > 0$ and $g \in \mathbb{R}$,*

$$\operatorname{spec}(H_{a,g}) = \{E \in \mathbb{R} : |\Delta(E)| \le 1\}, \qquad \operatorname{spec}(H_{a,g}) = \operatorname{spec}_{\mathrm{ac}}(H_{a,g}). \tag{7}$$

There is neither point nor singular-continuous spectrum. The full-line Bloch multiplicity is two in every band interior. If $g \neq 0$, each band edge is a simple periodic or antiperiodic fibre eigenvalue. If $g = 0$, the positive Bragg contacts $(n\pi/a)^2$, $n \ge 1$, are double fibre eigenvalues; $E = 0$ is the simple free bottom.

Proof. Equation (6) and unimodularity give (7). The fibre eigenvalue branches are nonconstant real-analytic functions, with local analytic relabelling at crossings. Their critical points are isolated, so pushing Lebesgue measure in θ through the branches is absolutely continuous; this is also the standard periodic-singular Floquet theorem. Thus their direct integral has neither point nor singular-continuous part. Interior energies have the two distinct angles $\pm\theta$. At an edge the fibre multiplicity is the dimension of $\ker(M \mp I)$. For $g \neq 0$, M cannot equal $\pm I$: at a Bragg point the jump factor is nontrivial, while away from a Bragg point the upper-right entry in (3) is nonzero. Hence the kernel is one-dimensional. For $g = 0$, free propagation equals $(-1)^n I$ at every positive Bragg contact, giving dimension two. The constant periodic mode is the sole zero-energy fibre eigenfunction. ☐

2 Negative spectrum and the zero threshold

Put

$$q = ga, \qquad z = Ea^2. \tag{8}$$

This scaling removes a from the atlas. For $g < 0$ write $h = -q > 0$ and $y = \kappa a$. Then

$$\Delta_-(y) - 1 = 2\sinh(y/2)[\sinh(y/2) - \frac{h}{2y}\cosh(y/2)], \tag{9}$$

$$\Delta_-(y) + 1 = 2\cosh(y/2)[\cosh(y/2) - \frac{h}{2y}\sinh(y/2)]. \tag{10}$$

The functions $2y\tanh(y/2)$ and $2y\coth(y/2)$ are strictly increasing; their ranges are respectively $(0, \infty)$ and $(4, \infty)$.

Theorem 3 (Complete negative and zero-energy atlas)**.** *If $g > 0$, there is no nonpositive spectrum. If $g = 0$, the spectrum begins at zero. Suppose $g < 0$, and let $y_+ > 0$ be the unique solution*

$$h = 2y_+ \tanh(y_+/2). \tag{11}$$

The first band has lower edge $-y_+^2/a^2$, where $\Delta = 1$.

1. *If $0 < h < 4$, zero lies in the interior of this band; its upper edge is positive; it is specified in Theorem 4.*
2. *If $h = 4$ (equivalently $ga = -4$), zero is its simple antiperiodic upper edge, with $\Delta(0) = -1$.*
3. *If $h > 4$, there is a unique $y_- > 0$ satisfying*

$$h = 2y_- \coth(y_-/2), \tag{12}$$

with $y_- < y_+$. The first band is exactly $[-y_+^2/a^2, -y_-^2/a^2]$, and $(-y_-^2/a^2, 0]$ lies in a gap.

Proof. For $g > 0$, (5) is greater than one at every nonpositive energy. For $g < 0$, equations (9) and (10) reduce the two edge equations to (11) and (12). The derivative of the first edge function is positive. After multiplication by $\sinh^2(y/2)$, the derivative of the second has numerator $\sinh y - y > 0$; its left limit is four. The factor signs show that $|\Delta_-| \le 1$ for $0 \le y \le y_+$ when $h \le 4$, and for $y_- \le y \le y_+$ when $h > 4$. Reversing $E = -y^2/a^2$ proves every case. ☐

3 Complete sign atlas and gap asymptotic

For $x = a\sqrt{E} > 0$ the two factorizations

$$\Delta - 1 = 2\sin(x/2)[\frac{q}{2x}\cos(x/2) - \sin(x/2)], \tag{13}$$

$$\Delta + 1 = 2\cos(x/2)[\cos(x/2) + \frac{q}{2x}\sin(x/2)] \tag{14}$$

show both the fixed edges $x = n\pi$ and their partners. A partner adjacent to $n\pi$ obeys the single parity-free equation

$$q = 2x_n \tan\frac{x_n - n\pi}{2}. \tag{15}$$

Its left-hand side as a function of x has derivative $(x + \sin(x - n\pi))/\cos^2((x - n\pi)/2) > 0$ on the relevant open cell.

Theorem 4 (Positive bands and all open gaps)**.** *The positive spectrum is as follows.*

1. *If $q > 0$, equation (15) has a unique $x_n \in (n\pi, (n+1)\pi)$ for every $n \geq 0$. The bands are*

$$B_n = [x_n^2/a^2, ((n+1)\pi/a)^2], \qquad n \geq 0,$$

and the gaps are $((n\pi/a)^2, x_n^2/a^2)$ for $n \geq 1$, together with the gap below B_0.

2. *If $q = 0$, $\mathrm{spec}(H_{a,0}) = [0, \infty)$; the folded free bands meet at every Bragg energy and no gap is open.*

3. *If $-4 < q < 0$, there is a unique $x_1 \in (0, \pi)$ and the first band is $[-y_+^2/a^2, x_1^2/a^2]$. If $q = -4$, set $x_1 = 0$ and the first band is $[-y_+^2/a^2, 0]$. If $q < -4$, there is no positive x_1 and the first band is the negative interval in Theorem 3. In all three cases, for every $n \geq 1$ there is a unique $x_{n+1} \in (n\pi, (n+1)\pi)$ and*

$$B_n = [(n\pi/a)^2, x_{n+1}^2/a^2].$$

For $-4 \leq q < 0$ the first positive gap is $(x_1^2/a^2, (\pi/a)^2)$. For $q < -4$ it is the single gap $(-y_-^2/a^2, (\pi/a)^2)$ crossing zero. All later gaps are $(x_n^2/a^2, (n\pi/a)^2)$, $n \geq 2$.

Consequently every Bragg gap is open for every $g \neq 0$.

Proof. The monotonicity following (15) gives uniqueness. On the right cell the edge function ranges from zero to $+\infty$; on a left cell it ranges from $-\infty$ to zero. The exceptional first left cell instead has limiting value -4 at zero. The signs in (13) and (14) then distinguish the allowed and forbidden intervals and give the displayed ordering. A nonzero q cannot solve (15) with $x_n = n\pi$, so none of the stated gaps collapses. ☐

Corollary 5 (Controlled high-energy widths)**.** *For fixed $q = ga \neq 0$, let G_n be the gap adjacent to $(n\pi/a)^2$ and not containing lower-energy exceptional structure. As $n \to \infty$,*

$$x_n - n\pi = \frac{q}{n\pi} - \frac{q^2 + q^3/12}{(n\pi)^3} + O_q(n^{-5}), \tag{16}$$

$$|G_n| = \frac{2|g|}{a} - \frac{\mathrm{sgn}(q)(q^2 + q^3/6)}{(n\pi)^2 a^2} + O_q(n^{-4}a^{-2}). \tag{17}$$

In particular, there are computable $n_0(q)$ and $C(q)$ for which the absolute remainder in (17) is at most $C(q)/(n^4a^2)$ for $n \geq n_0(q)$.

Proof. Put $N = n\pi$ and $x_n = N + \delta_n$. Equation (15) is $q = 2(N + \delta_n)\tan(\delta_n/2)$. With $u = N^{-1}$ and $\delta_n = uv$, the equation is analytic at $(u, v) = (0, q)$ and its v-derivative there is one. The analytic implicit-function theorem and Taylor expansion give $v = q - (q^2 + q^3/12)u^2 + O_q(u^4)$, proving (16) with a locally bounded Taylor remainder. Expanding $|(N + \delta_n)^2 - N^2|/a^2$ gives (17) and the stated bound. ☐

4 Integrated density of states and edge density

Enumerate the closed bands increasingly as $B_0, B_1, \ldots$, including the negative first band when $g < 0$. Their lower and upper discriminants are $(-1)^j$ and $(-1)^{j+1}$. On the interior of B_j define the unwrapped phase

$$K(E) = \frac{j\pi + \arccos((-1)^j\Delta(E))}{a}. \tag{18}$$

Theorem 6 (IDS and DOS with band indexing)**.** *The integrated density of states per unit length is zero below B_0, equals*

$$N(E) = \frac{K(E)}{\pi} = \frac{1}{a}[j + \frac{1}{\pi}\arccos((-1)^j\Delta(E))] \tag{19}$$

on B_j, and is $(j+1)/a$ in the gap following B_j. It is continuous at all edges. In every open band,

$$\rho(E) = N'(E) = \frac{|\Delta'(E)|}{\pi a\sqrt{1 - \Delta(E)^2}}. \tag{20}$$

For $E = k^2 > 0$,

$$\Delta'(E) = -\frac{a\sin(ka)}{2k} + \frac{g(ak\cos(ka) - \sin(ka))}{4k^3}, \tag{21}$$

with $\Delta'(0) = -a^2/2 - ga^3/12$. For $g \neq 0$ every finite edge has the usual one-sided inverse-square-root DOS singularity. For $g = 0$ the folded formula is interpreted continuously and reduces to $N(E) = \sqrt{E}/\pi$ and $\rho(E) = 1/(2\pi\sqrt{E})$ for $E > 0$.

Proof. Each fibre band supplies one state per cell as its Bloch angle traverses $[0, \pi]$. The orientation alternates, which is exactly corrected by $(-1)^j$ in (18). This proves (19) and the constant gap values. Differentiating yields (20); direct differentiation of (4) yields (21) and its Taylor limit. The edge equations and their strict monotonicity imply $\Delta'(E_*) \neq 0$ for $g \neq 0$, hence the square-root law. Direct simplification gives the free formulas. $\square$